



Internet Governance: Infrastructure and Institutions

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CHAPTER

2 2 Models of Internet governance

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Abstract

This chapter presents a typology of various models of governance forms that impinge, or are assumed to impinge, on the Internet. Five such models are analysed. One model is based on a view of the Internet as a self-governing realm of individual liberty beyond the reach of government control. Another model takes as its point of departure the inherently cross-border nature of the Internet and sees transnational, quasi-private cooperatives or international organizations based on treaty arrangements as the most relevant institutions for Internet governance. A third model is based on the notion that many regulatory decisions are made by the code and architecture of the Internet. A fourth model is premised on the idea that as the Internet grows in importance fundamental regulatory decisions will be made by national governments through legal regulation. Finally, there is the model of market regulation and economics which assumes that market forces drive the fundamental decisions about the nature of the Internet. The chapter argues that no single one of these models captures all of the facets of Internet governance or offers the solution to all of the problems raised by Internet governance. It concludes with a discussion of the issue of network neutrality and attempts thereby to cast light on the future utility of the models concerned.

Keywords: [cyberspace](#), [spontaneous ordering](#), [network neutrality](#), [international organizations](#), [code](#), [Internet architecture](#), [national governments](#), [market regulation](#)

Subject: [Political Economy](#), [Organizational Theory and Behaviour](#), [Information and Communication Technologies](#)

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2.1. What is Internet Governance?

Internet governance is a large, complex, and ambiguous topic. When we think about regulation of the Internet, we might be thinking about a narrow but important set of questions about specific institutions, such as the Internet Engineering Task Force¹ (IETF) or the Internet Corporation for Assigned Names and Numbers² (ICANN): these institutions can be said to *govern* the technical infrastructure and architecture of the Internet. We might also be thinking about a much broader and perhaps more compelling set of questions about policy issues that implicate the Internet: these questions include the regulation of online gambling, child pornography, freedom of speech, and the future of commerce and implicate nation states and international organizations. For the purposes of this investigation, Internet governance implicates both the narrow questions about Internet infrastructure or architecture and the broad questions about regulation of applications and content. Moreover, the broad and narrow questions are related. Regulation of the technical infrastructure of the Internet has implications for the regulation of applications and content.

p. 49

Begin with the Internet itself. The Internet is a global network of networks, with communication between networks enabled by a communications protocol suite, currently TCP/IP. The Internet enables computers or servers that are attached to the net to communicate with one another, creating a platform on which software applications can run. The application that is most commonly associated with the Internet is the World Wide Web, which utilizes HTTP. But the Internet is not identical to the World Wide Web and it provides a platform for a host of other applications, such as email, file transfer protocol, and a variety of peer-to-peer file sharing programs.

In the broad sense, the Internet is a complex entity that includes the hardware and software technical infrastructure, the applications, and the content that is communicated or generated using those applications. In the broad sense, the Internet includes millions of computers running a myriad applications generating, manipulating, and retrieving a vast array of information. More concretely, the Internet in the broad sense includes the personal computer used to write this chapter, fibreoptic cable, routers, servers, web-browsers, Google, Yahoo!, MySpace, YouTube, the online edition of *The New York Times*, millions of weblogs, tens of millions of Internet-enabled mobile phones, and billions of email messages. In the broad sense, the Internet interpenetrates a wide array of ordinary human life—commerce, communications, entertainment, intimate interpersonal relationships, and a host of other activities increasingly occur in and through the Internet.

If the topic of Internet governance were taken as the investigation of the regulation of all these activities when they took place on (or were significantly affected by) the Internet, then ‘Internet governance’ would be more or less equivalent to ‘law and politics’ at least in the ‘wired’ and ‘wireless’ (or more developed) nations. This definition of Internet governance is simply too broad and ill-defined to be useful for the purpose of this investigation. We need a narrower and more focused conception of Internet governance to gain real traction on the issues.

p. 50

What then is the ‘Internet’ for the purpose of investigating ‘Internet Governance’? The narrow answer to this question is that Internet governance is about the ordering of whatever technical systems enable the operation of the global network of networks as a platform for applications. The Internet has a history in which particular technical systems have come and gone; there was, for example, an Internet before TCP/IP,³ and it seems likely that the current version of TCP/IP will have a series of successors, within the existing structure of the Internet.⁴ Further, a new global network of networks may supplant the existing Internet—a development prefigured by Internet2.⁵ The fundamental object of a study of Internet governance is the general type—the universal network of networks—and not the specific token—the existing Internet, based on TCP/IP.

Nonetheless, the Internet as it exists today is important both for its own sake and as the most fully developed realization of a global network of networks. Today’s Internet is constituted by a set of technical systems, including TCP/IP, the system of IP numbers that identify individual computers or servers on the net, and the

Domain Name System (DNS) that provides alphanumeric equivalents of IP number addresses. Those technical systems enable the operation of a hardware layer—with routers, fibreoptic cable, individual servers, cellular systems, and even satellite links. The same technical systems enable the operation of an applications layer, ranging from the familiar (the World Wide Web, browsers, search engines) to the old and obscure (GOPHER) to novel and innovative applications that may become ‘the next big thing’ or fail to move outside the laboratory.

Thus, the narrow answer to the question ‘What is Internet governance?’ is regulation of Internet infrastructure, its current operation, and the processes by which it develops and changes over time. In other words, the narrow focus of Internet governance is about the processes, systems, and institutions that regulate things like TCP/IP, the Domain Name System, and IP numbers. These systems are fundamental—they determine the capacities of the Internet.

Yet the narrow answer to the what-is-Internet-governance question is not satisfactory on its own, because Internet governance in the narrow sense is connected in important ways to Internet regulation in the broad sense. The technical infrastructure of the Internet interacts with the ability of governments to regulate applications, content, and the human activities that are enabled and facilitated by use of the Internet. In other words, the technical infrastructure of the Internet is connected to the legal regulation of gambling and child pornography, to the efficiency and transparency of the world economic system, and to fundamental human rights, such as liberty of conscience and freedom of speech.

p. 51 The broad answer to the question ‘What is Internet governance?’ is that regulation of the Internet encompasses the policy questions that are *really different* when content and conduct are communicated and acted on and through the Internet. There is a real difference when the role of the Internet in a policy question is not merely accidental. This point is important. One function of the Internet is its role as a general system for communication between persons. The Internet has various modalities when it serves this role: email, voice, instant messaging, and so forth. The same content that is communicated via the Internet can also be communicated by phone, snail-mail, in-person conversations, teletype, and other media. Almost every imaginable policy problem—from the preservation of endangered species to the regulation of bid rigging—can involve Internet communications, but these problems are not *really different* when the only role of the Internet is to substitute for other communications media.

There are, however, a variety of regulatory problems where the architecture or structure of the Internet does make a real difference. Operationally, a real difference is present if the substantive analysis of the policy problem would be significantly altered if the next-best communications technology were substitute for the Internet. For example, in Internet commerce, the question arises whether consumers are bound by form contracts that may be formed by clicking ‘I accept’ in response to proposed terms (frequently in a scroll-down box)—sometimes called ‘click-wrap’ agreements.⁶ A similar issue arises in mail-order relationships, where the proposed terms may be in small print on the back of the order form. This appears to be a policy question where the Internet does not make a real difference: if a particular contract is binding in the mail-order context, it will be binding in the Internet context, and *vice versa*. On the other hand, there is a structural similarity between the problem of spam⁷ and the problem of junk mail, but in this case, the nature of the Internet arguably does make a difference for economic reasons (the costs of snail-mail are internalized to the sender to a greater degree than are the costs of email). Assuming the examples are correctly analysed, spam is a problem within the scope of an analysis of Internet governance, whereas ‘click-wrap’ agreements are not.

p. 52 A complete analysis of Internet governance requires that we address both narrow issues implicated by the institutions that govern the technical infrastructure and architecture of the Internet and the broader issues that are implicated by the ways in which the Internet transforms policy questions that directly implicate applications, communication, and conduct. Because the narrow issues are rarely of substantial intrinsic importance (architectural elegance of network design matters only to network engineers) and the broad issues are sometimes of great social importance (fundamental human rights matter to everyone), it is important that

investigations of Internet governance focus on the relationship between technical infrastructure and Internet architecture and the impact of the Internet on broad policy questions. This idea can be summarized as a slogan: *focus on the nexus between Internet architecture and social policy*.

2.2. Analytic Framework: Architecture, Policy Analysis, and Models

How can we parse the problems of Internet governance? The approach advanced in this chapter is based on three central ideas: (a) the idea that the Internet is constituted by its architecture or code; (b) the idea that the problems of Internet regulation can be analysed by using the conventional tools of policy analysis, including but not limited to: (i) normative theory, (ii) economics, and (iii) social choice theory; and (c) the idea that the logical space for discussing Internet governance can be captured via a set of ‘models’ or ideal types for Internet regulation.

Each of these three ideas is controversial and deserves further comment. The first idea is that architecture or code matters. This idea is challenged by the counter-thesis that there are no unique issues of Internet governance, because such issues are no different for the Internet than they are for other policy contexts. This challenge is prominently associated with Judge Frank Easterbrook of the United States Court of Appeal for the Seventh Circuit and formerly a prominent member of the law faculty of the University of Chicago. In an essay entitled ‘The Law of the Horse’, Easterbrook famously wrote:

[T]he best way to learn the law applicable to specialized endeavors is to study general rules. Lots of cases deal with sales of horses; others deal with people kicked by horses; still more deal with the licensing and racing of horses, or with the care veterinarians give to horses, or with prizes at horse shows. Any effort to collect these strands into a course on ‘The Law of the Horse’ is doomed to be shallow and to miss unifying principles. Teaching 100 percent of the cases on people kicked by horses will not convey the law of torts very well. Far better for most students— better, even, for those who plan to go into the horse trade—to take courses in property, torts, commercial transactions, and the like, adding to the diet of horse cases a smattering of transactions in cucumbers, cats, coal, and cribs. Only by putting the law of the horse in the context of broader rules about commercial endeavors could one really understand the law about horses.

Now you can see the meaning of my title. When asked to talk about ‘Property in Cyberspace,’ my immediate reaction was, ‘Isn’t this just the law of the horse?’ I don’t know much about cyberspace; what I do know will be outdated in five years (if not five months!); and my predictions about the direction of change are worthless, making any effort to tailor the law to the subject futile. And if I did know something about computer networks, all I could do in discussing ‘Property in Cyberspace’ would be to isolate the subject from the rest of the law of intellectual property, making the assessment weaker.⁸

Analogously, one might object to the very idea of an essay on ‘Internet Governance’. Internet Governance, it might be argued, should receive the same attention as ‘Ranch Governance’ or less flatteringly, ‘Stable Governance’. There is waste that needs to be shovelled, but it does not rise to the level of academic interest.

The premise of this chapter and the book in which it appears is that the Internet does pose distinct and important problems—the analysis of which is strengthened rather than weakened by paying close attention to the nature of the Internet and the interactions between Internet architecture and Internet regulation. This is brought out by a simple point that draws on Easterbrook’s analogy between Internet law and the law of the horse. Horses are commodities—the law regulates property rights in horses, contracts involving horses, and even races involving horses. But the law does not regulate the internal operation of horses and the physiology of horses does not significantly differ from that of other animals. If the law were to attempt to regulate horse

physiology directly or if horse physiology were radically different from that of other animals in a way that affected the question whether horses could be property, there would indeed be 'a law of the horse'. When it comes to the Internet, there are policy questions about architecture and architecture does impact upon other policy questions—as becomes evident in the discussion that follows. For this reason, the Easterbrookian thesis that a focus on Internet governance will make policy analysis worse rather than better is mistaken.

p. 54 Yet Easterbrook was not entirely wrong. The second idea upon which this chapter is premised is that Internet policy can be analysed using the conventional tools of policy analysis. This idea is challenged by ↵ the counter-thesis that Internet governance should be driven exclusively or primarily by considerations of network engineering. This idea is not clearly articulated in the Internet-governance literature, because that literature is almost always the product of what might be called Internet outsiders—those who are not focused on the technical problems of Internet architecture and design. From the vantage point of a network engineer, it might look as though the fundamental problem of Internet governance is to prevent economists, policy analysts, and lawyers from interfering with good network engineering. That is, good Internet governance institutions should empower network engineers and computer scientists to make design decisions on the basis of the functional imperatives of the Internet itself. At a very high level of abstraction, that means that the chief aim of Internet governance is to produce an efficient global network of networks: everyone should be able to connect irrespective of hardware or operating system; the network should be neutral with respect to content of information that is transported, and should accomplish these goals efficiently (minimizing lag and maximizing throughput). Concomitantly, politicians, executives, economists, and policy analysts who are ignorant of network architecture are likely to make bad decisions on such matters—accidentally imposing huge costs as the unintended consequences of some short-term external policy goal.

There is, of course, something to the idea that the imperatives of network design should play an important role in shaping Internet governance institutions. That notion will be explored in some detail in this chapter. Yet from the fact that architecture is relevant to Internet governance, it does not follow that architecture should be the sole or exclusive factor in Internet governance. The Internet is not simply a congeries of research projects; it is increasingly becoming the most important medium for communication of ideas and information. The architecture of the Internet has consequences going beyond the realm of values that are relevant to network engineers. For example, the Internet facilitates anonymous encrypted communication that can be either impossible or extremely difficult to intercept or prohibit.⁹ This information may be innocuous or it may be illegal sharing of copyrighted audio or video files; communications may be innocent and personal or they may be part of a terrorist plot. To the extent that Internet architecture facilitates ↵ or impedes regulation of content, Internet governance will have consequences that transcend engineering imperatives.

p. 55

In addition, Internet governance deals with decisions about resource allocation. Domain names and IP numbers are economically scarce resources; economists have relevant things to say about how such resources can best be distributed or traded. Internet architecture may have effects on innovation in applications: economists may have things to say about the value of such innovation and the question whether network neutrality will or will not have harmful consequences for the rate of innovation.

Similarly, the institutions of Internet governance implicate questions of institutional design and normative theory. For example, if an Internet governance institution were structured so as to require consensus among interested stakeholders before policy decisions could be made, public choice theory would predict hold-out problems and rent-seeking behaviour. Even if the participants in the process have technical backgrounds, governments and firms are unlikely to allow Internet governance to operate without a consideration of their interests when the stakes are high.

The third idea is that the complex problems of Internet governance can usefully be analysed by reference to *ideal types* or *models* of Internet governance. This idea is challenged by the counter-thesis that each problem of Internet governance is best understood contextually on the basis of particular problems and tailored solutions.

The best answer to this challenge is the most straightforward one: simply present the models and demonstrate that they can do interesting work. The next part of this chapter introduces five models: (a) the model of spontaneous ordering; (b) the model of transnational and international governance institutions; (c) the model of code; (d) the model of national regulation; and (e) the model of market-based ordering. The models do not reflect the positions of particular analysts, nor do they have exact analogues in existing institutions. Rather, these models are ideal types: we take a set of ideas about Internet governance and construct a model that is consistent with the logical implication of the ideas. Actual policy proposals and institutions inevitably reflect a mix of ideas and therefore are hybrids of the simplified models.

In sum, the framework of the chapter is based on the thesis that Internet governance is best understood by examining models of Internet governance in relationship to the interaction between Internet architecture and conventional policy analysis.

2.3. Five Models of Internet Governance

p. 56

The fundamental subject of Internet governance is institutional—what institutions, if any, shall govern the Internet and how shall they be organized. Even a quick survey of existing Internet governance mechanisms reveals a complex set of institutions. At the surface level, there are a variety of institutions that are specifically targeted at Internet governance—the IETF, ICANN, and various other organizations. In addition, governments play a role. All national governments assert the power to regulate the physical infrastructure of the Internet and the activities of end users and content providers that take place within national boundaries. Further, the architecture of the Internet or its ‘code’ can be seen as a kind of Internet governance institution, because architecture may influence the feasibility and cost of various regulatory alternatives. Moreover, even after the ‘dotcom’ bust, it is apparent that the Internet is regulated by the market, and current controversies over network neutrality provide further evidence that market forces can impinge on the fundamentals of network design.¹⁰ Although some have argued that the best (or even inherent) design for Internet governance is ‘spontaneous ordering’ or a separate realm called ‘cyberspace’, the best view is that Internet governance is a hybrid of various models.

The sections that follow explore five models of Internet governance:

- The model of cyberspace and spontaneous ordering which is premised on the idea that the Internet is a self-governing realm of individual liberty, beyond the reach of government control.
- The model of transnational institutions and international organizations which is based on the notion that Internet governance inherently transcends national borders and hence that the most appropriate institutions are transnational quasi-private cooperatives or international organizations based on treaty arrangements between national governments.
- The model of code and Internet architecture which is based on the notion that many regulatory decisions are made by the communications protocols and other software that determine how the Internet operates.
- The model of national governments and law which is based on the idea that as the Internet grows in importance fundamental regulatory decisions will be made by national governments through legal regulation.
- The model of market regulation and economics which assumes that market forces drive the fundamental decisions about the nature of the Internet.

p. 57

2.4. Cyberspace and Spontaneous Ordering

Early thinking about the Internet is strongly associated with a romantic conception of cyberspace as a separate realm outside of physical space and the reach of either national governments or market forces. For example, David Post and David Johnson famously wrote:

Many of the jurisdictional and substantive quandaries raised by border-crossing electronic communications could be resolved by one simple principle: conceiving of Cyberspace as a distinct 'place' for purposes of legal analysis by recognizing a legally significant border between Cyberspace and the 'real world'. Using this new approach, we would no longer ask the unanswerable question 'where' in the geographical world a Net-based transaction occurred. Instead, the more salient questions become: What procedures are best suited to the often unique characteristics of this new place and the expectations of those who are engaged in various activities there? What mechanisms exist or need to be developed to determine the content of those rules and the mechanisms by which they can be enforced? Answers to these questions will permit the development of rules better suited to the new phenomena in question, more likely to be made by those who understand and participate in those phenomena, and more likely to be enforced by means that the new global communications media make available and effective.¹¹

Johnson and Post imagined that cyberspace could not and should not be regulated by national governments:

Governments cannot stop electronic communications from coming across their borders, even if they want to do so. Nor can they credibly claim a right to regulate the Net based on supposed local harms caused by activities that originate outside their borders and that travel electronically to many different nations. One nation's legal institutions should not monopolize rule-making for the entire Net. Even so, established authorities will likely continue to claim that they must analyze and regulate the new online phenomena in terms of some physical locations. After all, they argue, the people engaged in online communications still inhabit the material world, and local legal authorities must have authority to remedy the problems created in the physical world by those acting on the Net. The rise of responsible law-making institutions within Cyberspace, however, will weigh heavily against arguments that describe the Net as 'lawless' and thus connect regulation of online trade to physical jurisdictions. As noted, sysops, acting alone or collectively, have the power to banish those who commit wrongful acts online. Thus, for online activities that minimally affect the vital interests of sovereigns, the self-regulating structures of Cyberspace seem better suited to dealing with the Net's legal issues.¹²

When Johnson and Post were writing in the mid-1990s, this utopian vision of cyberspace as a separate realm beyond the reach of national governments may have seemed credible. Although they wrote only a little over 10 years ago, neither governments nor markets had fully digested the importance of the Internet and traces of a frontier atmosphere characterized by consensual transactions between individuals were still prominent. Certainly, the legal and economic environment of 2008, when this chapter was finalized, is different. It is now difficult to conceive of cyberspace as a sort of 'independent country'. Governments and large multinational firms now have pervasive presences in cyberspace.¹³

Nonetheless, there is a kernel of important truth in the model of cyberspace and spontaneous ordering. The architecture of the Internet is resistant to purely national control. Because the Internet is a global network of networks capable of transmitting any information that can be digitalized, it would be costly for any national government to attempt to comprehensively monitor all of the content on the Internet inside its national boundaries. Data on the Internet cannot be intercepted with the same relative ease as telephone calls can be monitored. On the Internet, data are broken into packets which may travel various routes to their final

destination; the telephone system is line switched, meaning that a telephone call can be intercepted at any point in the telephone network. Routing on the Internet is very flexible. If a government tries to block one computer or server, data can be rerouted via what is called a 'proxy server'.¹⁴

p. 59 We are all familiar with the consequences of the high cost of government regulation of the Internet. Content that once could be forbidden is now available. For example, local and state governments in the United States could effectively limit access to pornography before the Internet; after the Internet, the costs of such regulation would be astronomically ↵ high. Although China is able to make access to some content relatively difficult and expensive, sophisticated Internet users in China are able to circumvent, at least in part, the so-called 'great firewall'.¹⁵ The Internet may not be a separate, self-governing, libertarian utopia, but it is still a realm that hampers government regulation.

2.5. Transnational Institutions and International Organizations

Closely related to the idea that cyberspace is an independent realm outside the control of national governments is the notion that the Internet should be governed by special transnational institutions that are outside the control of national governments and instead answer to the 'Internet community' or the 'community of network engineers'. In direct competition with this model is a rival that accepts the premise that the Internet is inherently international in character, but rejects the premise that it should be outside the control of national governments. This rival view would replace special transnational institutions—like ICANN or the IETF—with international organizations on the model of the International Telecommunications Union (ITU) or the World Intellectual Property Organization (WIPO).¹⁶ Although we are examining these two ideas in tandem, they represent opposing tendencies—we are really dealing with two models here and not one. The core idea that both models share is that Internet governance requires institutional structures that cross national boundaries. The central disagreement between the two models is over the role of national governments: ICANN and the IETF are not international organizations created by treaties; WIPO and the ITU are agencies of the United Nations. So far, Internet governance institutions have followed the model of special transnational institutions, although there has been considerable pressure to move towards an international organizations model.¹⁷

p. 60 ICANN can serve as a case study of the model of special transnational institutions. As elaborated in Chapters 3 and 5, ICANN has an organizational structure that is almost baroque in its complexity. This is partly a function of ICANN's institutional history. It will be recalled from Chapter 1 that ICANN replaced a predecessor organization, the IANA, which was essentially run by one individual, Jon Postel—himself an important Internet pioneer. In the Internet's early history, Internet governance by ↵ personal authority was both possible and efficient—there was no need for an international bureaucracy for what was a cooperative research project governed by consensual decision-making of network engineers. By the time Postel was replaced by ICANN, the reality had changed and the Domain Name System implicated the interests of millions of users, multinational corporations, and national governments. Concomitantly, ICANN's complex organizational structure reflects the broad constellation of interests presently affected by domain name policy. Registries and registrars for the gTLDs have a direct economic stake in domain name policy, as do the operators of the ccTLDs. Because of the phenomenon of cybersquatting, the owners of trademarks also have a significant economic stake. Internet Service Providers (ISPs) and network administrators also have interests that are affected by ICANN.

Despite the complexity of its structure, at its heart ICANN is a non-profit corporation with a self-selecting and self-perpetuating board of directors. In economic terms, ICANN is a firm. It is certainly not a government or a treaty-based international organization. Admittedly, it has a special relationship with the US Department of Commerce. When ICANN was founded, responsibility within the US Government for the Internet had been transferred to the Department of Commerce (DOC); hence, that agency was involved in the process that led to the creation of ICANN in 1998. The precise legal nature of the relationship between DOC and ICANN is murky,¹⁸

but whatever that relationship may be, ICANN is far from being a mere arm of DOC or the US government more generally.

The bylaws of ICANN contain a statement of core values that give pride of place to ensuring the ‘operational stability, reliability, security, and global interoperability of the Internet’¹⁹—technical imperatives that reflect the corporation’s origins in the community of network engineers. There is an emphasis too on making ‘well-informed decisions based on expert advice’—another indicator that ICANN is attempting to preserve the ideal of cooperation between engineers in the face of pressure to both democratize policymaking and to allow for the operation of competitive market mechanisms.

p. 61 Despite the nods to democratic participation and competition, ICANN is best understood as an attempt to institutionalize and preserve the autonomy of an engineering-based approach to Internet governance in face of pressures to cede control to either national governments or ↴ market mechanisms. Two fundamental facts confirm this hypothesis. First, ICANN has remained as an independent, non-governmental organization despite efforts by international organizations such as the ITU to establish influence over the Domain Name System.²⁰ Second, the corporation has for a long time resisted market approaches to the creation of new top-level domains. That is, ICANN represents a distinct model of Internet governance.

The model of international organizations has never established substantial authority to engage in Internet governance. The ITU and other suborganizations of the United Nations are interested in various aspects of Internet policy: for example, Internationalized Domain Names (which allow use of non-Roman character sets) have received attention from the UN Asia-Pacific Development Information Programme.²¹ Yet to date, the role of international organizations in Internet governance has been limited to input and has not included formal authority.²²

2.6. Code and Internet Architecture

p. 62 The fundamental idea behind the model of code and Internet architecture can be called the *code thesis*.²³ The code thesis is the claim that the nature of Internet or cyberspace is determined by the code—the software and hardware that implements the Internet.²⁴ Unlike natural environments and processes, *how* the Internet runs or cyberspace operates is dependent on the code that implements it. This is a point at which the model of code and Internet architecture departs from the model of cyberspace and spontaneous ordering. The latter model—at least in simple forms—assumes that cyberspace has an inherent nature, and that regulation of the Internet must respond to that nature. The contrast between the two models is reflected in their different attitudes towards the question whether the Internet can be regulated by national governments. The model of cyberspace and spontaneous ordering is based on the premise that the Internet cannot be regulated by national governments, because, ↴ as a global network of networks, activity on the Internet can originate in any physical location, anywhere in the world.²⁵ The model of code and Internet architecture can concede that there is a large measure of truth in this claim, but supplements that concession with the claim that the Internet has this property because of the code or software that makes physical location irrelevant. That code could have been different or it could change in the future.

In this sense, code is the prime regulator in cyberspace—in Larry Lessig’s felicitous phrasing, ‘the Code is Law’.²⁶ The core of Lessig’s point is that software or code has regulative effects on human behaviour. In this sense, Internet architecture is like the architecture of buildings and cities. Just as the architecture of a building enables and encourages humans to move and congregate in certain ways, so the architecture of the Internet enables some activities by users and regulators while discouraging others.

What then is the architecture of the Internet? More pointedly, which feature or features of Internet architecture are responsible for the distinctive and valuable functionality that the Internet provides? Lessig suggests that

the answer to these questions can be captured in large part by a single principle; he has argued that the primary characteristic of the Internet architecture that enables innovation is *the end-to-end principle*.²⁷ As Lessig explains, the end-to-end principle says to keep intelligence in a network at the ends or in the applications, leaving the network itself to be relative simple. In short, the principle calls for 'stupid network' and 'smart applications'. The network simply forwards or routes the data packets and does not—and cannot by architecture—discriminate or differentiate traffic generated by different applications. The software at the transport and Internet protocol layers simply does not include code that would allow the Internet to associate data packets with application file types.²⁸ Thus, the Internet does not 'know' whether a given packet of data is a webpage (utilizing HTTP), a social science article downloaded from the Social Science Research Network²⁹ (utilizing FTP), an email message (utilizing Simple Message Transfer Protocol, SMTP), or an MP3 file being shared by use of, say, the peer-to-peer file-sharing program KaZaa. Nor does the Internet 'know' when email is being used to send a large file as an attachment. As a consequence, the Internet cannot coordinate the routing of packets so that they all arrive at a destination computer at the same time. Ninety-five per cent of the packets may arrive in the first minute, and the remaining five per cent, travelling by a different route on the Internet, might arrive minutes later. The Internet cannot decide that delivery of files that contain academic articles is a high priority for an academic institution and delivery of MP3 files for students is a low priority.

This lack of awareness illustrates what is sometimes called the 'stupidity' of the Internet. Alternatively, we can call this characteristic of the Internet *transparency*. The Internet is *transparent* to applications. The transparent, non-discriminatory nature of the Internet, Lessig has argued, is what has enabled the explosion of innovation and creativity on the Internet.³⁰ Since the network is non-discriminatory or transparent to applications and the intelligence is implemented at the ends by the applications, the innovation or creative activity is placed in the hands of the application creators. Thus, innovation is decentralized and the opportunity to devise new applications is available to millions of creative individuals with Internet access.

The economic significance of transparency is that it dramatically lowers the investment required to produce a given innovation. Given a transparent Internet, an innovator need only invest in innovation at the application layer. If the Internet were opaque, then new applications would also require investment in changes at lower layers (the transport, IP, link, or physical layers).

Moreover, transparency reduces the cost of innovation to consumers. Given a transparent Internet, a consumer need only invest in an application itself in order to make use of it. If the Internet were opaque, consumers would be required to invest in changes to their network infrastructure in order to utilize an innovative application. Because networks are complex and used by multiple users, network reconfiguration may be relatively expensive. Because the Internet is even more complex and used by hundreds of millions of servers, reconfiguration of the whole Internet to enable a new application would involve a very large investment.

This point about transparency and adoption costs leads to another important concept, *networking effects*. The economic value of some innovations depends on networking effects.³¹ The value of an application like SMTP (email) is a function, in part, of the number of adopters; the more users of email, the more valuable it is.³² For some applications, there may be a tipping point, at which the number of adopters reaches critical mass resulting in a discontinuous and large increase in value from networking effects or from similar phenomena with respect to awareness of (or belief in) networking effects. Once this tipping point is reached, the application becomes sufficiently valuable to broaden its appeal from early adopters to ordinary users. Reducing the cost of adoption for consumers increases the likelihood that these networking-effect gains will be realized. If adoption costs are too high for early adopters, the tipping point may never be reached. For this reason, transparency may be required to enable innovations that would be efficient even given the large costs associated with reconfiguring the Internet. Without transparency, the benefits of the innovation would never become apparent and hence the investments would never be made.

The economics of Internet innovation are illustrated by development of the World Wide Web by Tim Berners-Lee.³³ The core idea of the Web is the hypertext link, by clicking here, on this word, you go there, to that word. By clicking here, on this link, you go there to that webpage. Although Berners-Lee developed this idea independently, a similar idea had arisen previously, on several occasions—and before the rise of the Internet.³⁴ What the transparency of the Internet enabled was the implementation of Berners-Lee's Hyper Text Transfer Protocol (HTTP) and Hyper Text Markup Language (HTML) applications without any need for cooperation from or modification of the communications system on which they were implemented. That is, the transparent nature of the Internet dramatically lowered the investment that Berners-Lee and others had to make in order to produce the Web.

p. 65 The World Wide Web sits on top of the Internet. From the application user's point of view, the Internet is simply invisible—except, of course, when network congestion calls it to our attention. If the architecture of the Internet had been opaque, then the counterfactual equivalent of TCP/IP would have required modification for HTTP/HTML to run on the Internet. And if that had been the case, there is good reason to believe ↪ that there would never have been a World Wide Web. The networking effects that transformed the World Wide Web from a merely great idea into an enormously useful communications tool would never have begun cascading if the platform on which the Web runs had not already been widely accessible. Berners-Lee faced real obstacles in getting HTTP/HTML accepted, even though he was giving these away and they could be run for free (at least from the point of view of short-run marginal costs).³⁵ Had Berners-Lee been required to convince managers of the Internet to expend resources to enable the Internet to run HTTP/HTML, we have every reason to believe that he would have failed. Indeed, he might have been unable to convince his own employers to allow him to make the necessary modifications in the network at CERN for the experimental work that developed HTTP/HTML.

The economics of Internet transparency and networking effects are very powerful. HTTP/HTML enabled what we now think of as 'the Internet revolution'. But the 'dotcom' boom and bust, eBay and Amazon.com, homepages and browsing—the cyberspace phenomena that have become the furniture of our day-to-day ordinary existence—all of these are the product of one innovative application suite, HTTP/HTML. The peer-to-peer phenomenon (from Napster to KaZaa and beyond) is the product of a small number of innovative applications. Whatever the numbers, it is clear that the contribution of the Internet to global economic growth and development is substantial.

The model of code and Internet architecture can be interpreted in a second way that contemplates and reinforces the code thesis and the end-to-end principle. Transparency is not a direct result of the end-to-end principle, but rather is a built-in characteristic of the layered architecture of the Internet. That is, it is *layers* that are the key, central characteristic of the Internet architecture. The end-to-end principle emerged from the layers model as an articulation and abstraction of implicit ideas inherent in the layers model.

What are the layers of the Internet? Viewed as a system of communication between users, the six layers that constitute the Internet are as follows:

- the Content Layer—the symbols and images that are communicated;
- the Application Layer—the programs that use the Internet (e.g. the Web);
- the Transport Layer—TCP, which breaks the data into packets;
- p. 66 ↪ the Internet Protocol Layer—IP, which handles the flow of data over the network;
- the Link Layer—the interface between users' computers and the physical layer;
- the Physical Layer—the copper wire, optical cable, satellite links, etc.

The layers are organized in a vertical hierarchy. When information is communicated via the Internet, the information flows down from the content layer (the ‘highest’ level) through the application, transport, IP, and link layers to the physical layer (the ‘lowest’ level), across the physical layer in packets, and then flows back up through the same layers in reverse order. More specifically, communication on the Internet requires that content be digitalized by an application, and that the digital information be broken into packets by the transport layer and addressed by the Internet protocol layer so that it can be passed on by the link layer to the physical layer. Having reached the bottom layer, information then moves horizontally. The physical layer transmits the individual data packets by copper, fibre, and/or radio by various waypoints to an endpoint or destination on the network. Once at its destination, the information then ascends vertically through the layers to be interpreted by an application as content. In a nutshell, the fundamental architecture of the Internet is layered.

What is the relationship of end-to-end and layered architecture? The end-to-end principle derives from pioneering work in the theory of network design by Jerome Saltzer, David Reed, and David Clark. As they put it, end-to-end is an argument (or a class of arguments) that says:

The function in question can completely and correctly be implemented only with the knowledge and help of the application standing at the end points of the communication system. Therefore, providing that questioned function as a feature of the communication system itself is not possible.³⁶

Stated in a generalized form, the end-to-end argument counsels against low-level function implementation.³⁷ What does this mean? Avoiding low-level function implementation means that a functionality desired by an application should be implemented at the level of the application rather than at a lower level. The version of end-to-end quoted above is a statement of the end-to-end principle for the engineering of the Internet and the software that uses the Internet.

p. 67 For example, the functionality of ensuring data integrity for a file transfer application is better implemented at the application level than at the network communication level. Even if the network delivered data without corruption, the reasoning goes, the file transfer application still needs to do the data checking because the data corruption could happen before the data reach the network—due to a bad disk, for example.

The simple example of error checking illustrates two general features of the end-to-end argument—vertical hierarchy and information mismatch. Each of these two features must be explicated and clarified in order to illuminate the meaning and significance of the end-to-end principle in relation to the layers model.

The end-to-end argument presupposes some sort of vertical hierarchy— that is, it assumes lower and upper levels. In the above example, the file transfer application is at an upper level with respect to the network communication system, which, in turn, lies at a lower level with respect to the application. The vertical hierarchy in the above example is illustrated in Figure 2.1.

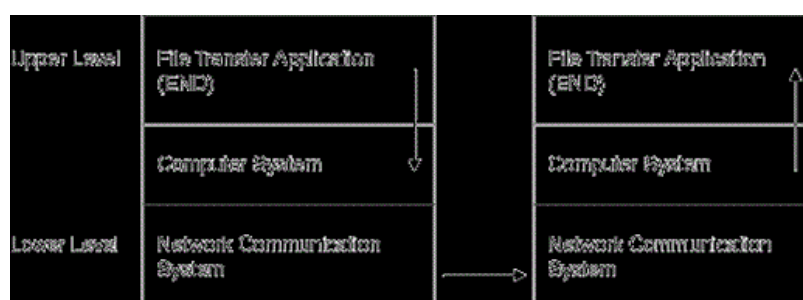


Figure 2.1. Vertical hierarchy in layered architecture

The fundamental insight of the end-to-end principle is that any given functionality should be implemented at a level where it is needed—the ‘end’ point. Thus, an essential task of understanding and applying the end-to-end principle is identifying and understanding the relevant levels. Where is the ‘end’? ‘End’ with respect to what? In the context of Internet architecture, the relevant vertical hierarchy implicit in the end-to-end argument is the layers of the TCP/IP protocol.

p. 68 In general, a network protocol is a set of rules and conventions by which computers can communicate with each other.³⁸ Since a computer ↪ network is nothing but a set of computers connected with each other, the network architecture is determined by the architecture of the network protocol.³⁹ The Transmission Control Protocol/Internet Protocol (TCP/IP) suite is the network communication protocol for the Internet. Thus, the architecture of the Internet as a network is determined by the architecture of the TCP/IP protocol.

The fundamental goal of the initial Internet architecture was to create a network of networks by interconnecting various computer network systems already in existence at the time.⁴⁰ In order to meet this goal, TCP/IP was designed to be software-only protocol, independent of any particular computer and network hardware. That is, TCP/IP is ‘pure code’. It is *the* code of the Internet that determines the architecture of the Internet. In a significant sense, then, TCP/IP is the most important institution of Internet governance.

2.7. National Governments and Law

The model of national regulation and law is premised on the idea that the importance of the Internet entails that it should be subject to regulation on the same basis and for the same reasons that other human activities are regulated. And, of course, Internet-related activity is subject to regulation. Examples abound. Publication on the web is subject to laws on defamation. Contracts made on the web are enforceable. Internet fraud is subject to criminal sanction. Peer-to-peer sharing of files with copyrighted materials subjects users to civil and/or criminal liability. It is important not to lose sight of the basic reality that most Internet regulation is regulation by national governments of Internet-related activities. That regulation establishes the conditions for Internet-related markets and enables the background conditions that enable other modalities of Internet regulation. In this sense, the model of national governments and law is triumphant— even if its presence is taken for granted as part of the background.

p. 69 The more difficult questions of Internet regulation concerning the application of national law arise in two paradigmatic contexts. The first context is the attempt to subject Internet architecture to national regulation. The second context is the attempt to censor open-access content. ↪ The two contexts are sometimes related—as the regulation of architecture may be a means to the regulation of content. In both contexts, national regulation creates substantial costs that caution against the extension of national law.

The first context where caution is appropriate is the direct legal regulation of Internet architecture. Interestingly, no national government has made a serious attempt to try to change the fundamental architecture of the Internet’s code. Arguably, any such attempt would be doomed to failure, because no one nation state has the regulatory clout to force a worldwide change in Internet architecture, and a successful change at the national level would result in the fragmentation of the Internet. However, the People’s Republic of China has instituted policies that aim at creating the functional equivalent of fundamental architectural changes.

China’s Internet regulations give the Chinese government a monopoly over all Internet connections going in and out of the country.⁴¹ The regulations designate the Ministry of Information Industry (MII) as gatekeeper to the Internet, and access to the global Internet by networks (ISPs) is restricted exclusively to a handful of channels—the national backbones— provided or sanctioned by the MII. Under the ‘Computer Information Network and Internet Security, Protection and Management Regulations’, access to certain objectionable

materials over the Internet is prohibited in China. The list of prohibited materials includes those (a) subversive of state power or the socialist system; (b) damaging to national unity; (c) inciting discrimination between nationalities; (d) disturbing to social order; (e) propagating feudal superstition; (f) related to pornography, gambling, or violence; (g) insulting or libellous; and (h) violating the Constitution or other laws. Pursuant to these regulations, a variety of international media outlets are blocked.

p. 70

Blocking of websites is delegated to the ISPs, with the Public Security Bureau (PSB) sending out a list of websites to be blocked from users in China. The ISPs are required by law to follow the directions from the PSB, and essentially operate as agents for the government. Technically, the ISPs in China are required to block all traffic from or to specific IP addresses handed down by the government. Since the Chinese government has a monopoly over the physical connection to the global Internet, and all ISPs that connect to the government sanctioned backbones are required to be licensed and thus required to follow the government orders, the government can completely block all objectionable contents from all of China, at least theoretically. It is like building an electronic Great Wall around China—hence the oft-used epithet ‘Great Firewall’.⁴²

However, uses of circumventing technologies—such as the peer-to-peer applications connected to anonymizing proxy servers outside China— make it difficult to completely block access based on the IP address of the true or ultimate destination. Nevertheless, in contrast to physical access control, China is deadly serious about enforcement of its IP layer regulation. First, rather than entirely relying on the ISPs to faithfully block banned sites, the Chinese authorities monitor and filter all Internet traffic going through China’s eight primary gateways to the global Internet. Presumably, the packets from the banned IP addresses are dropped at these backbone gateways. At the other end of the ISPs, installation of site-blocking software is required on all end-user computers with public access, such as those in Internet cafés. In 2002, for instance, Shanghai police closed down almost 200 Internet cafés in the city during a week- long sweep for not blocking sites as required under the law.⁴³ Similarly, Chinese authorities reportedly shut down 17,000 Internet bars that failed to install the site-blocking software.⁴⁴ Although closing down the service and confiscating the computers is directed at the physical establishment, it should properly be regarded as a part of IP layer regulation, as the nature of action is enforcement of the regulations at the IP layer—that is, blocking access to sites at specific IP addresses. Of course, the ultimate goal of the regulation is to control social or political unrest by restricting flow of information in and out of China.

p. 71

China’s regulation of the Internet illustrates the conflict between the model of national regulation and law and the model of code and Internet architecture. The Internet is not designed to facilitate national regulation—for example, the IP number system is not designed to create a hierarchical IP address space that is rigidly associated with national boundaries. Nor is the system designed to allow for the identification of content on the basis IP numbers. To the Internet, all IP addresses look alike so far as geography and content is concerned. As a result, China’s attempts to regulate content are costly and, at least with respect to sophisticated Internet users, ineffective.

The attempt to regulate Internet content at the national level is costly, because most of the materials on the blocked websites would be beneficial or desirable to millions of users within China, and would have been matters of little or no concern even to the government. The regulation clearly impairs the overall transparency of the Internet significantly by preventing access to popular Internet services from a very large region for a very large number of people. China’s attempt to regulate the Internet is ineffective because the Internet’s fundamental architecture facilitates circumvention of the Great Firewall: sophisticated users can use proxy servers or a virtual private network (VPN) to evade the blocking of particular IP addresses.⁴⁵

The second context for caution is the legal regulation of access to content by end users. In the case of China, this strategy would have been unavailing, because China lacked sufficient de facto power to influence content providers at an acceptable cost. Internet content providers outside of China did not have sufficient assets subject to Chinese control to give the Chinese government the means to coerce compliance with Chinese law

restricting content. But when the government that wants to control content does have sufficient de facto power, the outcome may be different. This possibility is illustrated by the French Yahoo! case.

The case arose in early 2000 as a consequence of Yahoo!'s auction site, which is accessible from its French portal.⁴⁶ Among the items offered for sale on Yahoo!'s auction site is war memorabilia, including—at least back in 2000—items connected to the Nazi party. A French court attempted to prevent Yahoo! from auctioning Nazi items to or from France. That is, it was attempting to control the content delivered by the Internet to users who accessed the Internet from within French territory.

p. 72 The French action resulted from suits filed by two French organizations, La Ligue Contre Le Racisme Et L'Antisemitisme (League Against Racism and Antisemitism, LICRA) and L'Union Des Étudiants Juifs De France (French Union of Jewish Students). These organizations brought a legal action against Yahoo! in the Tribunal de Grande Instance de Paris (the 'French Court'). The French petitioners claimed that the sale of Nazi-related goods through Yahoo!'s auction site that is accessible in France violated Section R645-1 of the French Criminal Code, which prohibits exhibition of Nazi propaganda and artefacts for sale.⁴⁷ Because any French citizen is able to access the Nazi-related materials on the web site, the French Court found that Yahoo!'s auction site violates the French law, and entered an order requiring Yahoo! to block French citizens' access to the areas of Yahoo!'s auction site that offer for sale any Nazi-related materials.⁴⁸

Yahoo! asked the French Court to reconsider its order, claiming that it was technologically impossible to selectively block French citizens from its auction site. Relying on expert testimony, however, the French Court found that blocking access from French citizens is technologically feasible. The panel of experts, including Vinton Cerf—one of the central designers of the Internet⁴⁹—testified that some 70 per cent of the IP addresses of French users or users in French territory could be correctly identified to be located within the French territory.⁵⁰ By combining the IP address method with voluntary identification or registration, the experts opined that Yahoo! can filter French users with a success rate approaching 90 per cent. The French Court thus denied Yahoo!'s request and reaffirmed its earlier order.⁵¹

p. 73 In response to the French Court's ruling, Yahoo! filed a complaint against the French organizations in United States District Court for the Northern District of California, seeking a declaratory judgment that the French Court's orders are neither cognizable nor enforceable under the laws of the United States. The defendants immediately moved to dismiss on the basis that the district court lacked personal jurisdiction over them. That motion was denied.⁵² Yahoo! then moved for summary judgment on its declaratory judgment claim.

When considering the case on its merits, the District Court first noted that the case was not about the right of France or any other nation to determine its own law and social policies.⁵³ 'A basic function of a sovereign state is to determine by law what forms of speech and conduct are acceptable within its borders,' said the Court.⁵⁴ The issue was, according to the Court, 'whether it is consistent with the Constitution and laws of the United States for another nation to regulate speech by a US resident within the United States on the basis that such speech can be accessed by Internet users in that nation.'⁵⁵

The conflict between the French and American judgments in the French Yahoo! case illustrates the inherent difficulty of national regulation of content on the Internet. Because the Internet is a global network of networks, the only way to ensure that content does not cross national boundaries is to eliminate the content altogether—meaning that the law of one nation will affect the availability of content throughout the world. The United States District Court in the French Yahoo! case based its decision on the principle that US law should trump the law of France in this situation. The District Court reasoned that 'although France has the sovereign right to regulate what speech is permissible in France, this Court may not enforce a foreign order that violates the protections of the United States Constitution by chilling protected speech that occurs ... within our borders'.⁵⁶ Accordingly, Yahoo!'s motion for summary judgment was granted.

The French Yahoo! case exemplifies the central difficulty with the model of national regulation. That model assumes that the Internet can be regulated in accordance with the principle that each sovereign has power over its own territory. Taking the experts' opinion at its face value, it is possible to identify the geographical location of the computers on the Internet on the basis of their IP addresses with about 70 per cent accuracy. Thus, the blocking ordered by the French court would block about 70 per cent of the users from France, as well as many outside the country. The real problem, however, is what the figure 70 per cent represents. First of all, the experts themselves emphasized that 'there is no evidence to suggest ↵ that the same will apply in the future.'⁵⁷ In other words, the figure has no predictive value. The 70 per cent is a statistical figure estimated from the accidental history of how IP addresses were allocated around the world.⁵⁸ Moreover, it is a figure that was specific to France. It is not based on architecture or design constraints of the Internet. And there is nothing in the architecture of the Internet to prevent reassignment or reallocation of blocks of IP addresses to a different region or country. Thus, the 70 per cent given by the experts is not a stable figure that can be a basis for lasting regulation or policy. Therefore, any territorial blocking based on IP addresses is inherently over-inclusive and under-inclusive. How severe the problem is may be anyone's guess at a particular point in time.

p. 74

The second problem is that IP address-based blocking can be easily circumvented by readily available technologies such as anonymizers. With such technology, users connect to a site through another server that hides the true origin of the user, and determination of the geographical location of the user based on IP address is consequently made impossible.⁵⁹ Thus, the anonymizers can render the regulation ordered by the French Court entirely under-inclusive where it really counts. That is, those users in France who insist on purchasing the Nazi-related materials on the Yahoo! site will easily circumvent the IP-based blocking, and the regulation would have very little impact.

There are technologies that can be used to 'dynamically discover' geographical location of the hosts based on their IP addresses.⁶⁰ But these technologies involve some sort of guessing game—for example, guessing physical distance by network response delay—that cannot guarantee a high rate of success.⁶¹ In fact, under the current Internet architecture, any effort to map IP addresses to geographical location is fundamentally bound to be an inexact science, although rough and ready identification may be possible.⁶² This is because allocation and distribution of IP ↵ addresses are fundamentally activities above the TCP/IP layers—addresses are assigned without any *architectural* constraints or mandates—and information about them is simply not communicated to the IP layer. Thus, all efforts to map IP addresses to geographical location have the inherent problems of being over-inclusive and under-inclusive.

p. 75

National regulation of the Internet does work in a variety of contexts, particularly when all of the parties to the regulated activity are within the physical territory of a particular nation state. But national regulation of the Internet is inherently costly and ineffective when the object of regulation is either the architecture of the Internet or content that originates outside of national boundaries. Thus, the model of national regulation by law cannot provide a complete solution to the problems of Internet governance.

2.8. The Market and Economics

If national regulation does not represent a complete solution to the problems of Internet governance, what about the market? By describing the problems of Internet governance as problems of regulation, an implicit assumption has been made—that market mechanisms do not provide a solution that obviates the need for government intervention or the creation of special transnational institutions. The model of the market and economics is premised on the idea that this implicit assumption is false. Rather than view Internet governance as a regulatory problem, the model attempts to redescribe the underlying phenomena in economic terms, as markets for products and services.

The economic approach to Internet governance can be illustrated by returning to ICANN and its regulation of the Domain Name System (DNS). There is much to be learned by stepping back and looking at the DNS as an ordinary service, provided by an ordinary organization, subject to the familiar laws of supply and demand. How can the provision of that service be organized so as to provide the greatest benefit of the public? How can the DNS be put to its highest and best use? As elaborated in Chapter 5 (Section 5.1), at the heart of the DNS is the root directory—the part of the system that allows the creation and utilization of top-level domains, like .com, .edu, and .org.

p. 76 Root service is a scarce resource in the *economic sense*. The term ‘scarce’ can be used in a different sense, which we call the *engineering sense*. It is important to contrast and compare these two different senses of scarcity. When network engineers or network administrators approach the question of scarcity of domain names, they bring a particular perspective to the table. Does my name server have sufficient capacity to provide name service for everyone who is on my network? Is the set of allowable names large enough, so that I can give a name to everyone who makes a request? Thus, the network administrator for Widgets, Inc., would ask whether her name server can support all the third-level domains (3LDs) that are in use or likely to be in use. Perhaps, she needs to support www.widgets.com, ftp.widgets.com, and network.widgets.com. From her point of view, there is no scarcity. Even if she needed to add hundreds or thousands of 3LDs to her zone file, her name server could handle the work. No matter how many requests for 3LDs she gets, she will not run out of character strings. From her point of view, there is no scarcity in the SLD name server system or in the 3LD name space.

We can look at scarcity in the root in exactly the same way that our hypothetical engineer looks at scarcity in Widget’s SLD. From the network engineer or network administrator’s perspective, the root is not a scarce resource. There are currently less than 300 top-level domains (TLDs). This number does not even come close to taxing the capacity of the root server system. Moreover, if a very large number of new gTLDs did begin to tax the capacity of the system, it could simply be re-engineered. The theoretical capacity of the name space is vast.⁶³ From the engineering perspective, the root is not a scarce resource.

There is, however, an economic sense of the word scarcity that differs from the engineering sense. Economic scarcity exists whenever something is costly, even if it is abundant. A network engineer might say that there is no scarcity of capacity on an Ethernet network if the engineer has planned for sufficient resources (optical fibre, etc.) to meet anticipated demand into the foreseeable future. The same engineer might say that storage space on the email server is scarce, if the server is reaching its physical limit. To the economist, both resources are scarce. The various components that produce capacity on an Ethernet network all have costs. The optical fibre that created practical abundance is not free.

p. 77 The root is a scarce resource in the economic sense for two distinct and independent reasons. First, the root server system itself is economically ↵ scarce. Second, the name space is economically scarce. If either of these propositions is true, then root service is a scarce resource from the economic point of view.

The root server system is scarce in the economic sense, because root service is not free. To provide root service, there must be root servers (computers that provide root service to the Internet). The root servers (i.e. the computers) are not free.⁶⁴ The software that runs the root servers is not free—it took human labour to create that software. The root servers must be maintained, and the labour that does the maintenance is not free. Root service is scarce in the economic sense.

This same point can be made by examining the capacity of the root server system. The precise capacity of the root server system is not documented and probably the upper limit has not been tested. Assume that a conservative upper limit on the current capacity of the system is from 1,000 to 10,000 TLDs. There is demand for TLDs, and the supply is, at least temporarily, limited by the capacity of the root system. It follows from the definition of economic scarcity, that TLDs are scarce and, therefore, if TLDs were sold, they will command a price.

So far this discussion has assumed a static root server system, but the hardware and software providing root service are not fixed in stone; therefore, the potential upper limit on the supply of TLDs is not inherently static. Even if the existing root server system can only support thousands of TLDs, a new and improved root server system might be able to support millions or even billions of TLDs. But this does not change the fact that root service is a scarce resource in the economic sense. Upgrading the root server system would be costly, and this fact suffices to establish the conclusion that root service is scarce in the economic sense.

There is another reason why the root is a scarce resource. The root name space is itself scarce. Names are simply strings of characters. If there were no upper limit on the length of a name string, then the name space as a logical construct would be infinite. Even if the limit were 10, the number of logically possible names would be in the quadrillions—so many that there are no conceivable circumstances under which the supply would be exhausted.⁶⁵ Nonetheless, the name space is a scarce economic resource. The economic scarcity of the root flows from the differential value of different names. For example, .com is more valuable than .kjd-7xx9-a, and this is true despite the fact that either string will do equally well at the job of connecting an end user to a server on the net.

There are two reasons to believe that different names have different values. The first reason is that the market prices different names differently—as elaborated in Chapter 5. The second reason is that different names have different useful characteristics; another way of putting this second point is to say that different names have different functional utilities.

Consider the first point. Experience in secondary SLD markets demonstrates that different names have different economic values. And there is little reason to doubt that the same would apply for TLDs if there were a market. Imagine, for example that ICANN conducted an auction for the following two TLD strings: 'sex' and '8ki3—d'. It is obvious that the former TLD string would command a relatively high price,⁶⁶ and the other string would either attract no bids or bids at a much lower level.⁶⁷

There is a good reason why the market prices different strings differently. Not all names are created equal. Due to their semantic and typographical properties, some names are more useful than others. For example, some names are more memorable than others. Strings that are memorable, such as .biz or .web, are more valuable than strings that are difficult to remember.

From the standpoint of economics, a good place to begin an analysis of root allocation and the creation of TLDs is the question, 'Should root service be provided by the market?' One reason for answering such a question in the negative is that the good or service in question is a 'public good' in the sense of a good that ought to be provided by government or a public entity. National defence and clean air are usually considered public goods. Conversely, if root service is a 'private good', without significant negative externalities (or social costs), then well-established and uncontroversial economic theory suggests that it can best be provided by markets.⁶⁸

The phrase 'public good' is ambiguous. In one sense, the public good is simply whatever is in the interest of the public as a whole; in this sense, 'public good' is a synonym for 'common weal'. Economists use the phrase 'public good' in a more restricted and technical sense. A given good or service is a 'pure public good' if and only if it meets the following two criteria: (a) non-rival consumption and (b) non-excludability. By 'non-rival consumption', economists mean that consumption of the good by one individual does not limit the availability of the good to any other individual. By 'non-excludability', economists mean that the providers of the good or service are unable to exclude individuals from access to the good.⁶⁹

Is root service a public good? In the domain name policy literature, statements are sometimes made suggesting that the answer to this question is yes. For example, the operating principles of the Government Advisory Committee to ICANN state that 'the Internet naming and addressing system is a public resource that must be managed in the interests of the global Internet community'.⁷⁰ Because the notion of a 'public resource' is not defined, it is difficult to discern what this statement means.

Root service is clearly not a public good in the economic sense. Although root service might look like it is unlimited and free from a user's perspective, the same is not true from the perspective of a provider or proprietor of root service. First, root service to the proprietors of TLD registries is rivalrous. If the root points to a name server (or system of parallel name servers) operated by Verisign to provide name service for the .com TLD, then it cannot also point to a name server (or system) operated by a different registry for the .com TLD. For name service to work, each domain name must resolve to a unique IP address. This in turn requires that each second-level domain must be identified by a unique (or coordinated set of) first-level domain name server(s). If Verisign operates the name server for .com, then no one else can operate that same name server for a given root. Hence, root service, as well as other names services, is rivalrous in the economic sense.

p. 80 Second, root service to the proprietors of TLD registries is excludable. Any given TLD can be either included or excluded from the root. Thus, the operator of the root can sell root service to TLD registries. If the proprietor of a given TLD registry refuses to pay, then the operator of the root can simply eliminate the TLD from the root or point to a name server operated by a rival registry proprietor. Hence, root service is excludable in the economic sense. By extension, all levels of name service are similarly excludable.

What are the implications of the conclusion that root service is a private and not a public good? At this stage, we will set aside the question whether there can be competition for root service. Given that root service is a private good, it could be provided efficiently by profit-seeking entities, such as for-profit corporations.

Private firms' sale of root service could be accomplished through a variety of pricing mechanisms. As discussed above, it would be difficult to sell root service directly to users and therefore root service providers would be more likely to sell root service to the proprietors of TLDs. Root service would be one of the factors (costs) of operating a TLD, and hence would be incorporated into the price the TLD operators charge to registrants of SLDs, assuming the TLD proprietor was in the SLD business.⁷¹ What price would be charged? If there were competition in the market for root service, the price would equal the costs of root service (including, of course, the cost of capital in the form of interest to lenders and dividends or share price appreciation for equity holders). Given that the cost of providing root service is relatively low,⁷² the price would be low.

Why should private goods be provided by markets? At a fundamental level, the answer to this question lies in the Pareto Principle:⁷³ given the possibility of improving the welfare of one individual without harming anyone, the Pareto Principle requires that we take the action that results in the improvement. A market transaction, where individual X and individual Y voluntarily exchange some good or service for payment, is required by the Pareto Principle, unless the transaction results in an externality—that is, a cost or harm to some third party.

p. 81 Markets both allow Pareto-efficient transactions and give incentives for all such transactions to take place. In a market, the purchaser who derives the greatest benefit from a good or service will be willing to pay the highest price. For this reason, markets put resources to their highest and best use.

In the context of domain name policy, the point is that a market will allow the root to be put to its highest and best economic use. Unless a particular TLD would create a negative externality—a harm to the root itself or Internet users—the Pareto Principle says that TLD should be created. This is the economic version of what has been called *Cerf's Principle*: a string should enter the root if it will do no harm.⁷⁴ Economists would extend this principle: a TLD string or slot should be freely exchangeable, unless its exchangeability would harm some third party.

The idea that name service could be viewed as a private good does not imply the further conclusion that name service should be free of government regulation on the basis of social costs (or 'externalities' in the economic sense). For example, the string of characters that make up a name could make a false and injurious statement, triggering liability for defamation. Because domain names can communicate brief messages, they can become the vehicle for the creation of social costs similar to those created by other communications media. In this

regard, domain names might be viewed as similar to text messages—but with the important additional feature that domain names are ‘texted’ to the whole world instead of a particular recipient.

The most important example of social costs associated with domain name registrations is ‘cybersquatting’—the registration of a domain name that infringes a trademark for the sole purpose of extracting a payment for transfer of the domain name. If the root were to be auctioned, this would create the opportunity for ‘root cybersquatting’.

Of course, if a new TLD were used to sell a product or service in a manner that infringed a trademark, the operator of the TLD would be subject to civil liability as would anyone who used any communications medium. That a new medium of communication creates new opportunities for trademark infringement is hardly an argument for treating the medium as a public good. If that were the case, webpages should be treated as public goods, since any webpage used to sell a good or service can potentially be used for trademark infringement.

Cybersquatting differs from ordinary trademark infringement because cybersquatters do not use domain names to sell products or services. The purpose of cybersquatting is to extract a payment from the owner of the trademark. From the point of view of the economic model of Internet governance, the question is whether cybersquatting is inefficient or—to use slightly different terminology—whether cybersquatting creates a dead weight loss.

Economic analysis of this question can begin with the Coase Theorem— named for the Nobel Prize winning economist Ronald Coase.⁷⁵ Coase observed that in the absence of transaction costs, the assignment of entitlements does not affect the efficiency of outcomes. As applied to the problem of cybersquatting, the idea is very simple. If cybersquatters and trademark owners could bargain with zero costs, then domain names which had their highest and best use as the property of the trademark owner would all be sold to the trademark owners. Of course, there would be a distributional effect (a transfer from trademark owners to cybersquatters), but this does not result in economic inefficiency.

In the real world, however, the transfer of trademarked domain names from cybersquatters to copyright owners would result in positive transaction costs. Trademarked domain names are not commodities that have standard prices. Would the root name ‘.ibm’ be worth a thousand euros or a million euros? It is not difficult to imagine that the negotiations to determine the answer to that question would consume considerable resources. Given positive transaction costs, the allocation of the original entitlement can be demonstrated to affect the efficiency of the outcome. In the case of domain names, this suggests that the right to register a trademarked name should be allocated to the trademark owner. That result could be accomplished by regulations enacted by national governments or it could be instituted unilaterally by the proprietor of the Domain Name System in order to avoid the cost and trouble that could result if the registration of trademarked domain names involves the operator of the name system in litigation.

If we assume that the problem of harmful domain names can be handled by a combination of market forces and government regulation, the economic model faces another important question: is the provision of root service a natural monopoly? From the technological point of view, there is no barrier to the creation of alternative roots. ICANN could operate one root system, and one or more other entities could operate competing roots. As elaborated in Chapter 5 (Section 5.1), alternative root services do exist but these provide service to only a tiny fraction of Internet users. Why? The following paragraphs argue that the economics of root service strongly favour a single root. The argument does not depend on technical considerations. Even assuming that multiple roots were technically feasible, the marketplace would result in a single root (or perhaps a dominant root, with a tiny fraction of purchased domain names residing in alternative roots). This conclusion is based on two factors: (a) the economic notion of networking effects; and (b) the cost structure of root service providers. These two factors would inevitably lead a system that began with competing alternative roots to

evolve into a system with a single root. The same factors prevent a system with a single root from evolving into a system with multiple roots.

First, consider ‘networking effects’ or ‘networking externalities’—terms here being used in a technical economic sense.⁷⁶ The value of root service increases with the number of users of the service. A single root with many users is more valuable than a single root with few users. Given any arbitrary number of users, root service is more valuable if all of the users patronize the same root, and as a consequence, root service is less valuable if the same users divide their patronage among two or more competing roots. Second, consider the cost structure of root service providers. Much of the cost of operating a root is fixed. Although a root with more customers in theory requires more server capacity, given the distributed nature of the DNS, the marginal costs of serving additional customers are relatively small as a share of total costs. Given these two factors (networking effects and a high ratio of fixed to marginal costs), rational TLD proprietors will choose to purchase root service from the market leader. This is because the market leader provides a more valuable service at a lower cost. ISPs will point towards the market leader’s root server, because the market leader’s root service is more valuable to the ISP’s customers than is the root service provided by other root service providers. Both the networking and cost effects become more and more pronounced as the market share of the market leader becomes larger and larger. For this reason we would expect the market to lead to a single firm providing root service, that is, root service will be provided by a monopolist.

The way that networking effects operate with respect to root service can be illustrated by performing a thought experiment. Imagine that the root became fragmented. If there were many competing roots, and many competing sets of TLDs, then different users would reach different destinations by entering the same domain name. ICANN’s root presumably would result in www.amazon.com being resolved to the IP Address of the e-commerce retailer in Seattle, Washington. An alternate root might result in the same domain name being resolved to a tour operator in Brazil. Different Internet Service Providers (ISPs) would point requests for root service to different root servers. As a result, individuals would get different results for the same domain name when they moved from one ISP (at home) to another (at work or at an Internet café). As fragmentation increases, the value of domain names decreases.⁷⁷ If I am an Internet user, I am less likely to invest in memorizing or memorializing ‘www.amazon.com’ if it does not reliably get me to the website I am seeking. If I am a website proprietor, I am less likely to invest in publicizing a domain name, if users will frequently be directed to another proprietor’s website when they enter the string of characters that I am advertising.

Profit-maximizing firms with monopolies will (in the absence of price regulation) extract monopoly rents. That is, they will charge rents that exceed their costs. Unless the monopolist can successfully engage in price discrimination, monopoly rents will be inefficient. The monopoly rent raises the price of the monopoly good, and as a result, some consumers of the good who would have paid the market price will not consume the good. In the case of domain names, however, price discrimination is a real possibility.⁷⁸ For example, the monopoly proprietor of the root could auction TLDs.⁷⁹ If a firm wished to become the proprietor of the .biz TLD, the monopolist could auction .biz. In cases where no bidder possesses monopsony power,⁸⁰ the auction price should provide a monopoly rent to the holder of the root-service monopoly and also ensure that the TLD goes to its highest and best economic use.

There are, moreover, limits on the rent that a monopoly proprietor of the root could charge. Most obviously, the monopolist could not charge a rent in excess of the value of the TLD to potential purchasers. There are additional limits. At bottom, the Internet is a communications system, and as such, it competes with other systems. If the proprietor of a root charged too much for root service, end users and information providers would use alternative communications systems—for example, telephone, broadcast, and direct mail.

So far, we have assumed that the proprietor of the root is a profit-maximizing firm and will charge monopoly rents. Does ICANN seek to maximize profit? An adequate account of the institutional economics of non-profit corporations is far beyond the scope of this chapter. Nonetheless, it is fairly clear that ICANN does not

maximize profits in the way that for-profit firms do. That is, ICANN does not seek to maximize its revenues from root service. Indeed, ICANN continues to provide root service to ccTLD operators who refuse to pay ICANN any fee at all. ICANN, as a non-profit corporation, operates the root in the public interest. As a nonprofit corporation organized under the laws of the State of California, it must have a 'public or charitable purpose'.⁸¹

The obligation to act in the public interest is, however, abstract and vague. This is true both as a matter of political philosophy and as a matter of law. As a matter of political philosophy, the nature of the public interest is, at the very least, contested. That is, whatever the ultimate resolution of philosophical debates about what counts as the public interest, as a practical matter this is a question of which we are unlikely to see a strong social consensus in a modern pluralist democracy.⁸² The lack of consensus on the nature of the public interest is even more evident in the international context: the Internet affects diverse cultures with different value systems. As a matter of law, the obligation to act in the public interest underdetermines ICANN's actions.

p. 86 Undoubtedly, ICANN could ↵ make DNS policy in a variety of ways without endangering its status under Californian law, but there is one thing that ICANN cannot do given its legal status. ICANN cannot operate the DNS so as to maximize its own profits or so as to confer a private benefit on the various stakeholders that participate in ICANN operations.

What are the lessons of the economics of root service for the market model of Internet governance? It seems that the implications are mixed. On the one hand, economic analysis of the root suggests that ICANN's decision-making process regarding root expansion may have been—at least up until very recently—fundamentally flawed. Prior to June 2008, ICANN pursued a relatively restrictive policy on root expansion, resulting in artificial scarcity of TLDs and an inefficient allocation of those TLDs that have been created. However, this situation should change with ICANN's decision, announced in June 2008, to radically liberalize its policy.⁸³ On the other hand, networking externalities and the resulting monopoly suggest that competitive solutions may not be available for some services that are provided by Internet governance institutions. Thus, the basic premise of ICANN's creation—that name service should be provided in the public interest—may be valid, while ICANN's actual policy decisions prior to June 2008 may have been contrary to the public interest.

2.9. Hybrid Models

So which is the best model of Internet governance? The argument of this chapter has been that no single model provides the solution to all the problems that Internet regulation can address. Nonetheless, the analysis does suggest some tentative conclusions that can help to order thinking about making Internet governance policy.

First, a broad range of issues that involve the Internet can be resolved without recourse to any special institutions or principles of Internet governance. In a wide variety of circumstances, markets work perfectly well. When government is needed, many problems can be solved at the national level through regular lawmaking processes. In other words, the model of national regulation and the market model should be the first recourse of policymakers. It is only when markets and national ↵ governments both fail that special Internet governance institutions are required.

p. 87 Second, not all of the remaining issues of Internet regulation require solutions. In some cases, the best policy is to leave the Internet alone. For example, the Internet frustrates government policies that seek to inhibit freedom of speech and liberty of conscience. To use the old metaphor, this is a feature, not a bug. Although the romantic idea of the cyberspace as a self-governing realm of spontaneous ordering and inherent freedom is exaggerated, it does contain an important kernel of truth. There are advantages to those features of Internet architecture that frustrate national regulation. There should be no general presumption in favour of a system of either national or international regulation that would transform the Internet to facilitate legal control of content along national lines.

Third, there are great virtues to the existing system for Internet governance in the narrow sense—that is, the system for the regulation of and by code and architecture. That system has produced the transparent Internet that serves as an engine of innovation and open communication. The model of transnational governance institutions that supports the existing architecture for reasons of network engineering serves as an institutional bulwark for the preservation of these features of the Internet. The alternatives, national or international regulation, seem likely to degrade rather than enhance these desirable characteristics.

Fourth, the virtues of the current system do not entail the conclusion that improvements cannot be made. On the one hand, national regulation can become more sensitive to the value of preserving the Internet's architecture: in particular, national regulation should respect the end-to-end principle and the layered design of the TCP/IP. On the other hand, transnational Internet governance institutions can become more sensitive to the economics of Internet regulation and the possibility that an engineering perspective may lead to decisions that waste Internet as a resource.

These four conclusions lead naturally to a fifth. The best models of Internet governance are hybrids that incorporate some elements from all five models. Internet governance is a complex task requiring a complex set of regulatory mechanisms. As a result, the optimal system of governance is a combination of regulation by transnational institutions, respect for the architecture that creates transparency, national regulation, and markets.

p. 88

2.10. Internet Governance and the Future of the Internet: The Case of Network Neutrality

How will evolving models of Internet governance affect the future of the Internet? Current controversies over 'network neutrality' provide a case study for the complex ways in which modalities of Internet governance interact with substantive policy concerns.⁸⁴ Analysis of the issue might start with the model of Code. The architecture of the Internet is neutral among applications. TCP/IP is insensitive to the particular application; all packets are treated equally and no particular application or user is favoured over any other. As noted above, a variety of arguments can be made for network neutrality, importantly including the claim that the transparency of the Internet facilitates innovation.

The economic model of Internet governance might be used to generate a different set of insights. Network service providers can argue that they should be free to innovate in the provision of Internet service to firms and individuals. By discriminating against some services, such as peer-to-peer file sharing or streaming video, they may be able to provide a more responsive network environment. Consumers may prefer to purchase network services from a provider who discriminates against some packets in order to favour others. On the surface, at least, the economic model suggests that competition between service providers will provide the best answer to the question whether a neutral network or a discriminatory network is more valuable to consumers. Consumers who prefer a neutral network would patronize neutral ISPs; consumers who prefer a discriminatory network would buy broadband service from ISPs that discriminate against applications that 'hog' bandwidth.

p. 89

This simple answer to the network neutrality question is based on the assumption that there is efficient competition between providers of broadband service. There are questions about the competitiveness of the market for broadband services. Phone companies that provide DSL service⁸⁵ or cable television providers might have monopoly power—at least with respect to some consumers. Some consumers may not have a choice: DSL service is not available at all geographic locations. Even consumers who could theoretically choose between DSL and cable may face high switching costs and difficulty obtaining information about the network neutrality policies of their providers.

Some advocates of network neutrality argue that if broadband providers have monopoly power, they may attempt to leverage that power. For example, a cable provider might discriminate against Internet video in order to advantage their own streaming or 'on demand' video products. Standard economic models predict that this will not occur. Although the models themselves are technical, their core insight is easy to understand. The monopolist broadband providers can extract their full monopoly rent in the price they charge for broadband service. Once they are charging the full monopoly price, the attempt to extract even more by discriminating in favour of their own video products will simply result in less demand for their broadband services.⁸⁶ Other economists have challenged the conventional wisdom, arguing that increased profits in the market for the complimentary product (in this case, video) can exceed the loss in monopoly rents in the primary market (in this case, broadband services).⁸⁷ Most recently, Barbara van Schewick has demonstrated that monopolists in the provision of broadband services can have an incentive to discriminate against applications in a variety of circumstances. For example, if AOL had a monopoly position in the provision of instant messaging services and was a local monopolist in the provision of broadband service, it might have an incentive to exclude rival instant messaging services from its broadband network in order to preserve its monopoly in instant messaging. The Federal Communications Commission in the United States considered a similar argument when it approved AOL's merger with Time-Warner on the condition that the combined firm not exclude competitors' instant messaging services from their networks.⁸⁸ Van Schewick also demonstrates that even if the market for broadband services is competitive, broadband service providers may have incentives to engage in inefficient discrimination among applications.⁸⁹

p. 90 Even if we assume a competitive market and that broadband service providers will only discriminate against applications in response to consumer demand, such discrimination may generate social costs. As noted above in connection with the code model of Internet governance, network neutrality facilitates innovation at the application level. Neither consumers nor network service providers fully internalize these costs. An individual consumer's decision to patronize a discriminatory broadband provider will have a negligible effect on the benefits of application innovation to that consumer. Similarly, even a major broadband service provider will only absorb a small fraction of the total social costs of innovation that is discouraged by its discriminatory network. The magnitude of these costs is impossible to estimate in advance. Future application innovations may create economic value on the same order of magnitude as HTML created by enabling the World Wide Web, but it is also possible that there simply are no more 'killer applications' in the future of the Internet whether or not network neutrality is maintained.

The resolution of the economic arguments for and against broadband neutrality is beyond the scope of this chapter. The important point is that the economic model does not guarantee efficient results. This suggests that the network neutrality issue requires the engagement of another modality of Internet governance. The most obvious possibility is the model of national government regulation. In the United States, for example, the federal Congress has considered network neutrality legislation that would prohibit network service providers from discriminating among applications.⁹⁰

Of course, government regulation of network neutrality at the national level would need to be coordinated in order to preserve the value of transparent global Internet.

p. 91 Because regulation at the national level may be inadequate, it might be argued that the best solution to the network neutrality problem is provided by the model of transnational institutions and international organizations. As a practical matter, however, an international solution seems remote at best. None of the existing transnational governance institutions (such as IETF or ICANN) would conceive of network neutrality
↳ regulation as within their respective mandate. No existing treaty-based international organization has jurisdiction over the Internet, and no such organization would view domestic network neutrality as an appropriate topic for international regulation. Thus, an international solution to the network neutrality problem would likely require a new treaty regime, and any new treaty regime faces enormous political obstacles. Because the costs of a loss of global Internet transparency are 'invisible' and difficult to estimate, the

prospects for an international agreement that guarantees the future transparency of the global Internet seem dim at best.

What lessons for the future of Internet governance can be learned from the example of network neutrality? None of the models of Internet governance guarantees optimal policy. Transparency and neutrality were built into the TCP/IP, but the model of code requires the voluntary cooperation of all the key players to preserve neutrality between applications. In the early days of the Internet, when the key players were research institutions and cooperating government entities, there was some hope that the transparency built into the code could be preserved. This vision of Internet governance was extended to the earliest transnational Internet governance institutions—the IETF and IANA—but as the global Internet became economically important, giant corporations and national governments replaced universities and researchers in key roles. As a result, bottom—up, consensus-based decision-making that focuses on network engineering is no longer as feasible. Market forces and the interests that drive government regulation are sure to have a major role in shaping the future of the Internet. For this reason, the role of independent institutions and researchers that assess Internet policy and provide input into the decision-making processes of national regulators and transnational Internet governance institutions may be more important than ever. Multidisciplinary research that approaches problems of Internet regulation and governance from the perspectives of network engineering, law, economics, and social choice theory may offer the best hope for an Internet future that avoids catastrophic unintended consequences.

Notes

- 1 See further Chapter 4 (Section 4.1).
- 2 See further Chapter 3 (Section 3.2.8).
- 3 See Chapter 1.
- 4 See e.g. RFC 2460: Internet Protocol Version 6 (IPv6) (authors: S. Deering and R. Hinden) (December 1998), <<http://www.ietf.org/rfc/rfc2460.txt>>.
- 5 A consortium made up of a large number of US universities, working together with industry and government, to facilitate development and application of new Internet technologies. Further on Internet2, see <<http://www.internet2.edu/>>.
- 6 See, for example, Francis M. Buono and Jonathan A. Friedman, 'Maximizing the Enforce- ability of Click-Wrap Agreements', *Journal of Technology Law & Policy*, 1999, vol. 4, no. 3, <<http://grove.ufl.edu/~techlaw/vol4/issue3/friedman.html>>.
- 7 Further on spam and its regulation, see, for example, Credence E. Fogo, 'The Postman Always Rings 4,000 Times: New Approaches to Curb Spam', *John Marshall Journal of Computer & Information Law*, 2000, vol. 18, no. 4, 915–44.
- 8 Frank H. Easterbrook, 'Cyberspace and the Law of the Horse', *University of Chicago Legal Forum*, 1996, vol. 11, 207, 207–8. Contrast, for example, Lawrence Lessig, 'The Law of the Horse: What Cyberlaw Might Teach', *Harvard Law Review* 10.2307/1342331¹, 1999, vol. 113, 501–46.
- 9 See, for example, Chris Nicoll, 'Concealing and Revealing Identity on the Internet', in C. Nicoll, J. E. J. Prins, and M. J. M. van Dellen (eds.), *Digital Anonymity and the Law* (The Hague: T. M. C. Asser Press, 2003), 99–119; Ed Felten, 'Zfone Encrypts VoIP Calls', *Freedom to Tinker*, 23 May 2006, <<http://www.freedom-to-tinker.com/?p=1019>>, last accessed 30 April 2008.
- 10 See further Section 2.10.
- 11 David R. Johnson and David G. Post, 'Law and Borders—The Rise of Law in Cyberspace', *Stanford Law Review*, 1996, vol. 48, 1367, 1378–9.
- 12 David R. Johnson and David G. Post, 'Law and Borders—The Rise of Law in Cyberspace', *Stanford Law Review*, 1996, vol. 48, 1390–1.
- 13 See further Chapter 3 (Section 3.3).
- 14 Further on proxy servers, see <http://en.wikipedia.org/wiki/Proxy_server>.
- 15 See further Section 2.7 of this chapter and references cited therein.
- 16 For details of these organizations, see Chapter 3.
- 17 See further Chapter 6.
- 18 See further Chapter 3 (Section 3.2.8) and references cited therein.
- 19 The statement of core values is fully set out in Chapter 3 (Section 3.2.8).
- 20 International Telecommunications Union, *ITU and its Activities Related to Internet-Protocol (IP) Networks: Version 1.1* (April

- 2004), especially Section 3.2, <<http://www.itu.int/osg/spu/ip/itu-and-activities-related-to-ip-networks-version-1.pdf>>, last accessed 30 April 2008.
- 21 'Internationalized Domain Names', APDIP e-note 9/2006, <<http://www.apdip.net/apdipenote/9.pdf>>, last accessed 30 April 2008.
- 22 See further Chapters 3 and 6.
- 23 Lawrence Lessig, *Code, and Other Laws of Cyberspace* (New York: Basic Books, 1999).
- 24 Ibid. 6.
- 25 See, for example, Justin Hughes, 'The Internet and the Persistence of Law', *Boston College Law Review*, 2003, vol. 44, 359, 365 *et seq.*
- 26 Lessig, *Code*, 6.
- 27 Ibid. 34.
- 28 However, other software, for example, in the form of 'packet sniffers'—can examine the content of packets on a network. See further <<http://www.webopedia.com/TERM/s/sniffer.html>>. An example is Sniff'em, see <<http://www.sniff-em.com/>>, last accessed 30 April 2008.
- 29 See <<http://www.ssrn.com>>.
- 30 See Lessig, *Code*.
- 31 See generally Oz Shy, *The Economics of Network Industries* (Cambridge: Cambridge University Press, 2001).
- 32 For the purposes of illustration, we set aside the possibility that there may be a point at which additional users of email actually reduce its utility by creating sorting problems and encouraging spam.
- 33 See Chapter 1 (Section 1.12).
- 34 See Chapter 1 (Section 1.2).
- 35 See further Berners-Lee, *Weaving the Web*, especially chapters 2–5.
- 36 Jerome H. Saltzer, David P. Reed, and David D. Clark, 'End-to-end arguments in system design', *ACM Transactions on Computer Systems* 10.1145/357401.357402², 1984, vol. 2, 277, 278.
- 37 Ibid. 10.1145/357401.357402²
- 38 Andrew S. Tanenbaum, *Computer Networks* (Upper Saddle River, New Jersey: Prentice-Hall, 1996, 3rd edn.), 17.
- 39 Andrew S. Tanenbaum, *Computer Networks* (Upper Saddle River, New Jersey: Prentice-Hall, 1996, 3rd edn.), 18.
- 40 David D. Clark, 'The Design Philosophy of the DARPA Internet Protocols', *Computer Communication Review* 10.1145/52325.52336², August 1988, vol. 18, 106–14.
- 41 The ensuing description of Chinese regulation builds principally on the following sources: OpenNet Initiative Study, 'Internet Filtering in China 2004–2005', <<http://www.opennetinitiative.net/studies/china/>>, last accessed 30 April 2008; Philip Sohmen, 'Taming the Dragon: China's Efforts to Regulate the Internet', *Stanford Journal of East Asian Affairs*, 2001, vol. 1, 17–26; Ronald J. Deibert, John G. Palfrey, Rafal Rohozinski, and Jonathan Zittrain (eds.), *Access Denied: The Practice and Policy of Global Internet Filtering* (Cambridge, Massachusetts: MIT Press, 2008), 263–71.
- 42 See, for example, Jonathan Watts, 'Behind the Great Firewall', *The Guardian*, 9 February 2008, <<http://www.guardian.co.uk/technology/2008/feb/09/internet.china>>, last accessed 30 April 2008; James Fallows, 'The Connection Has Been Reset', *Atlantic Monthly*, March 2008, <<http://www.theatlantic.com/doc/200803/chinese-firewall>>, last accessed 30 April 2008.
- 43 'Shanghai Cracks Down on Net Cafes', *Reuters*, 6 May 2002, <<http://www.wired.com/news/politics/0,1283,52330,00.html>>, last accessed 30 April 2008.
- 44 Ibid.
- 45 See, for example, Fallows, 'The Connection Has Been Reset'.
- 46 For further detail on the case and its background, see, for example, Jack Goldsmith and Tim Wu, *Who Controls the Internet? Illusions of a Borderless World* (Oxford/New York: Oxford University Press, 2006), chapter 1.
- 47 Indeed, it would seem that the auctioning of Nazi material also contravened the rules posted by Yahoo! at that time for the operation of its auction site—thus weakening the legitimacy of Yahoo!'s initial stance. See further Joel R. Reidenberg, 'Yahoo and Democracy on the Internet', *Jurimetrics Journal*, 2002, vol. 42, 261, 265–6.
- 48 *LICRA v. Yahoo! Inc.*, T.G.I. (Tribunal de Grande Instance) Paris, 22 May 2000, No. 00/05308; unofficial English translation at <<http://www.lapres.net/yahen.html>>, last accessed 30 April 2008.
- 49 See Chapter 1 (Section 1.6).
- 50 'Experts testify in French Yahoo! Case over Nazi Memorabilia', CNN, 6 November 2000, <<http://www.cnn.com/2000/TECH/computing/11/06/france.yahoo.trial.ap>>, last accessed 30 April 2008.
- 51 *LICRA v. Yahoo! Inc.*, T.G.I. Paris, 20 November 2000, No. 00/05308; English translation at <<http://www.cdt.org/speech/international/20001120yahoofrance.pdf>>, last accessed 30 April 2008.
- 52 *Yahoo!, Inc. v. La Ligue Contre Le Racisme Et L'Antisemitisme*, 145 F.Supp.2d 1168 (2001).
- 53 Ibid. 1186.

54 Ibid. Hence, at least one court in the United States should have no problem with China's regulation of the Internet.
 55 Ibid.
 56 Ibid. 1192.
 57 *LICRA v. Yahoo! Inc.*, T.G.I. Paris, 20 November 2000, No. 00/05308 (English translation) at 9,
<http://www.cdt.org/speech/international/20001120yahoofrance.pdf>, last accessed 30 April 2008.
 58 Further on the IP address allocation scheme, see Chapter 5 (Section 5.1).
 59 *LICRA v. Yahoo! Inc.*, T.G.I. Paris, Nov. 20, 2000, No. 00/05308 (English translation) at 9.
 60 See, for example, Venkata N. Padmanabhan and Lakshminarayanan Subramanian, 'An Investigation of Geographic Mapping Techniques for Internet Hosts', *Proceedings of ACM SIGCOMM 2001*, 2001, August, 173–85,
<http://www.acm.org/sigs/sigcomm/sigcomm2001/p14-pabmanabhan.pdf>, last accessed 30 April 2008. 'ACM SIGCOMM' stands for Association for Computing Machinery Special Interest Group on Data Communication.
 61 Ibid. 178.
 62 See also Dan Jerker B. Svantesson, 'Geo-Location Technologies and Other Means of Placing Borders on the 'Borderless' Internet', *John Marshall Journal of Computer & Information Law*, 2004, vol. XXIII, 101–39, 111–21; Dan Jerker B. Svantesson, "Imagine there's no countries ..."—Geo-identification, the law and the not so borderless Internet', *Journal of Internet Law*, 2007, vol. 10, no. 9, 1, 18–21.
 63 See further Chapter 5 (Section 5.1).
 64 Of course, the owners or operators of the root servers may donate them to the system, but this does not make them free in the economic sense. Someone pays for the root servers. They do not appear out of the ether.
 65 The number of strings is $37^{10} + 37^9 + \dots + 37^1 = 4,942,156,160,540,570$ (roughly 5 quadrillion). That would currently allow nearly a million domain names per person on Earth.
 66 In US litigation in 2003, a federal judge valued the use of sex.com over a five-year period at US\$65 million: see *Kremen v. Cohen*, No. 01-15899 (9th Cir. 3 January 2003).
 67 Of course, there are imaginable scenarios in which the meaningless string would attract a high price. For example, if someone devised a use for a TLD that required that the string almost never be entered by accident, and if only one new TLD would be allowed into the root, an arbitrary, meaningless string might command a high price. But this example merely confirms the principal point, that different TLD name strings will command different prices in a market.
 68 Paul Samuelson, 'Pure Theory of Public Expenditure', *The Review of Economics and Statistics*, 1954, vol. 36, no. 4, 387–9. [10.2307/1925895](https://doi.org/10.2307/1925895)⁵¹
 69 See Dennis C. Mueller, *Public Choice II* (Cambridge: Cambridge University Press, 1989), 11; Mark Blaug, *Economic Theory in Retrospect* (Cambridge: Cambridge University Press, 1996, 5th edn.), 580.
 70 See Governmental Advisory Committee Operating Principles (adopted 25 May 1999),
<http://www.icann.org/committees/gac/operating-principles-25may99.htm>.
 71 Some TLD operators might not sell SLDs. For example, if .ibm were a TLD, then International Business Machines might use .ibm for its own purposes and not offer any other entities registrations in .ibm. In that case, the cost of .ibm would be passed on to the purchasers of IBM products.
 72 ICANN has provided the IANA functions for the entire Internet for only a few million dollars per year. See, for example, ICANN, *Proposed Fiscal Year 2004–2005 Budget* (17 May 2004), 7; available at <http://www.icann.org/financials/proposed-budget-14may04.pdf>, last accessed 30 April 2008.
 73 There are actually two Pareto Principles, Strong Pareto and Weak Pareto. Strong Pareto requires actions that improve the welfare of every person. Weak Pareto requires actions that improve the welfare of at least one person but do not make any person worse off. See, for example, Howard F. Chang, 'A Liberal Theory of Social Welfare: Fairness, Utility, and the Pareto Principle', *Yale Law Journal*, 2000, vol. 110, 173, 176–7.
 74 The principle is named after Vinton Cerf. It is a corollary of Karl Auerbach's 'First Law of the Internet,' which states: 'Every person shall be free to use the Internet in any way that is privately beneficial without being publicly detrimental. The burden of demonstrating public detriment shall be on those who wish to prevent the private use. Such a demonstration shall require clear and convincing evidence of public detriment. The public detriment must be of such degree and extent as to justify the suppression of the private activity.' See Karl Auerbach, 'First Law of the Internet' (27 March 2003),
<http://www.icannwatch.org/comments.pl?sid=1242&cid=11389>, last accessed 30 April 2008.
 75 Ronald H. Coase, 'The Problem of Social Cost', *Journal of Law and Economics*, 1960, vol. 3, 1–44 [10.1086/466560](https://doi.org/10.1086/466560)⁵².
 76 See, for example, Shy, *Economics of Network Industries*, 1–6.
 77 This conclusion would hold even if ISPs or third-party vendors provided a service that aggregated alternate roots, giving users a menu of the alternate destinations for a given domain name. The extra step, choosing among the alternative resolutions, increases the time required (and hence the cost) of the aggregated alternative root services.
 78 In fact, price discrimination already occurs with so-called premium domain names being sold for higher prices than 'standard' names. The former category of names is typically easy to remember and intuitively associated with potential e-

- commerce business models. See, for example, 'Premium .TV domain names', <<http://www.totalregistrations.com/services/twstvpop.html>>, last accessed 30 April 2008.
- 79 As already happens. See, for example, 'Dot-mobi domains go to highest bidders', *The Register*, 11 October 2007, <http://www.theregister.co.uk/2007/10/11/dotmobi_domain_auction/>, last accessed 30 April 2008.
- 80 There may well be cases in which one bidder will possess monopsony power. For example, the owner of a trademark may have a legal monopoly over use of the mark. For this reason, it may be that the proprietor of a TLD name that is identical or substantially similar to a trademark may have a legal right to exclude others from acting as proprietors of their TLD.
- 81 Californian Corporate Code § 5111.
- 82 Cf. John Rawls, *Political Liberalism* (New York: Columbia University Press, 1993).
- 83 See further Chapter 5 (Section 5.1.2).
- 84 See generally, Christopher S. Yoo, 'Network Neutrality and the Economics of Congestion', *Georgetown Law Journal*, 2006, vol. 94, 1847–1908; Christopher S. Yoo, 'Beyond Network Neutrality', *Harvard Journal of Law & Technology*, 2005, vol. 19, no. 1, <<http://jolt.law.harvard.edu/articles/pdf/v19/19HarvJLTech001.pdf>>; Zittrain, *The Future of the Internet*, 178–85; Thomas M. Lenard and Randolph J. May (eds.), *Net Neutrality or Net Neutering: Should Broadband Internet Services Be Regulated?* (New York: Springer Publishing, 2006); Barbara van Schewick, 'Towards an Economic Framework for Network Neutrality Regulation', *Journal of Telecommunications and High Technology Law*, 2007, vol. 5, 329–83.
- 85 Digital Subscriber Line, that is, transmission of digital data over a telephone network.
- 86 See, for example, Robert H. Bork, *The Antitrust Paradox: A Policy at War with Itself* (New York: Free Press, 1993, 2nd edn.), 372–5; Richard A. Posner, *Antitrust Law* (Chicago: University of Chicago Press, 2001, 2nd edn.), 198–9.
- 87 See Michael D. Whinston, 'Tying, Foreclosure, and Exclusion', *American Economic Review*, 1990, vol. 80, 840, 850–2.
- 88 Barbara van Schewick, 'Towards an Economic Framework for Net Neutrality Regulation', 352.
- 89 Ibid. 377.
- 90 See Declan McCullagh and Anne Broache, 'Republicans Defeat Net Neutrality Proposal', *CNET News.com*, 5 April 2006, <http://news.com.com/2100-1028_3-6058223.html>, last accessed 30 April 2008. For an overview of US government actions and statements of officials concerning network neutrality, see John Windhausen, Jr., 'Good Fences Make Bad Broadband: Preserving an Open Internet through Net Neutrality', (Public Knowledge White Paper, 6 February 2006), <<http://static.publicknowledge.org/pdf/pk-net-neutrality-whitep-20060206.pdf>>.